

Multilingual Long-Form Speech Instruction Following: KIT's Submission to IWSLT 2026

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Abstract

With the advent of Large Language Models, single-task and token-based multi-task models have evolved into instruction-based systems that infer task and target language implicitly from natural language prompts. This trend is reflected in IWSLT's Instruction Following Track, which this year introduced new tasks including an unknown surprise task, posing a genuine challenge against overfitting to known tasks. We present KIT's submission to the Long and Short Instruction Following tracks in the unconstrained setting. Our approach combines a general data augmentation pipeline that converts short-form corpora into long-form training data through segment concatenation, LLM-based label generation, and cross-lingual translation, yielding over 1M instances across six tasks and four languages. We further show that likelihood-based re-ranking, while highly effective for ASR, systematically degrades semantic tasks by spuriously selecting segmentation-based candidates, a failure mode resolved by combining likelihood with Minimum Bayes Risk decoding.

1 Introduction

Recent work has focused on integrating speech into LLMs to create Speech Language Models, typically by incorporating pre-trained audio encoders (Koneru et al., 2025; Züfle and Niehues, 2025; Tang et al.) into the LLM architecture. Alternative approaches train audio and vision encoders jointly with an LLM backbone, developing true multimodal Foundation Models (Xu et al., 2025a), with recent work also demonstrating the effectiveness of combining speech and vision modalities (Sinhamahapatra and Niehues, 2025; Koneru et al., 2026). Despite these advances, a significant gap remains: the effective processing of long-form audio (Papi et al., 2026). Most models rely on the Whisper encoder (Radford et al., 2022), which natively supports only 30 seconds per inference pass.

While newer models such as Phi-4 (Abdin et al., 2024) and Qwen2.5-Omni (Xu et al., 2025b) remove these architectural constraints, they lack exposure to long-form audio during training, a gap that manifests as significant performance degradation even on basic ASR (Papi et al., 2026). This paper presents KIT's submission to the [Unconstrained Long & Short Instruction Following tracks of IWSLT 2026](#). We participate in all four target languages: German, English, Italian, and Chinese. Our approach combines data augmentation to extend short-form datasets to long-form settings, temperature-scaled interleaving to balance task representation, and re-ranking to improve generation quality, exploring both end-to-end and cascaded architectures. The main contributions of this work are:

- A data augmentation framework that includes conversion of short-form speech datasets into long-form instruction-following training data through segment concatenation, LLM-based label generation, and cross-lingual reference translation, yielding over 1M training instances across six tasks and four languages from publicly available sources.
- Empirical confirmation that temperature-scaled data interleaving (T=2) consistently outperforms and converges faster than heuristic smoothed sampling in multimodal multi-task instruction following, extending prior findings from text-only settings to speech.
- A negative result on Chain-of-Thought task-token conditioning, showing that prefix-based task routing fails under training data imbalance and requires explicit balancing to be effective.
- A systematic comparison of six re-ranking strategies under the realistic constraint of no

task identity at inference time, revealing a previously uncharacterized failure mode where likelihood-based re-ranking spuriously selects segmentation-based candidates for semantic tasks.

- A combined Likelihood+MBR re-ranking strategy that resolves the ASR-vs-semantics tradeoff, achieving strong ASR improvement while limiting degradation on QA and summarization.

2 Data

This year’s IWSLT instruction-following task covered six tasks: Automatic Speech Recognition (ASR), Speech Translation (ST), Spoken Question Answering (SQA), Speech Summarization (SSUM), Audio Chaptering (ACHAP), and a surprise task. All tasks except ASR support the language pairs en–en, de, it, zh; however, not all task–language combinations have in-domain data readily available. A further challenge is data format: most existing datasets consist of short utterances under 30 seconds, while the long-form track requires audio up to 15 minutes. We address both gaps through a three-stage augmentation framework: (1) segment concatenation with speaker-aware grouping to construct long-form audio, (2) LLM-based label generation to create task annotations for unlabeled or partially annotated data, and (3) cross-lingual reference translation to extend English-only annotations to all target languages. The resulting corpus contains over 1M training instances spanning all six tasks and four language pairs, summarized in Table 1.

2.1 Translation Augmentation

Most datasets provide annotations only in English. To cover all target languages (de, it, zh), we translate English references using `translategemma-12b-it`¹ (Team et al., 2025).

We select this model based on strong reference-free translation quality (COMETKiwi (Rei et al., 2022)), outperforming alternatives such as SeamlessM4T-Large (Barrault et al., 2023) and LLaMA-3.1-8B-Instruct (Kassianik et al., 2025). This procedure is applied consistently across tasks for language coverage augmentation.

2.2 ASR

Our original ASR datasets consist of short audio-transcript pairs, typically under 15 seconds. However, the long-form track requires audio up to 15 minutes, necessitating enhancements to generate training data aligned with this evaluation setting.

YTSeg. We repurpose YTSeg (Retkowski and Waibel, 2024), a dataset originally curated for chaptering, as a source of long-form video for ASR. We retain videos up to 15 minutes, reducing the dataset from 16,404 to 10,729 examples (34.6% reduction). We further filter videos where Whisper Large achieves $\geq 50\%$ WER to remove noisy examples, yielding 10,638 examples (0.86% additional reduction). We then applied lightweight text normalization: removing non-lexical metadata such as background-noise markers in parentheses or square brackets (e.g., [music], (applause)), while preserving bracketed numeric content. We also normalized whitespace and excessive line breaks. This produced cleaner transcripts that better reflect spoken content for ASR training and evaluation.

NUTSHELL. We use NUTSHELL (Züfle et al., 2025), a dataset of ACL talk videos, which is particularly in-domain for our setting. Since NUTSHELL lacks ASR transcripts, we generate them using `parakeet-tdt-0.6b-v2`², applied to the audio track of each video. To align with the shared task’s 15-minute duration limit, we filter out videos exceeding this threshold.

EuroParl. We extend EuroParl-ST (Iranzo-Sánchez et al., 2020) for long-form ASR by concatenating aligned speech segments into 5–10 minute chunks. Segments are grouped by session and speaker, with additional speakers included if needed. Only transcribed audio portions are concatenated, excluding untranscribed gaps, yielding clean long-form audio-transcript pairs.

LibriSpeech. We extend LibriSpeech (Panayotov et al., 2015) by combining `train-clean-360` and `train-other-500` to increase acoustic diversity. Utterances are grouped by chapter and shuffled at the chapter level to mix conditions while preserving intra-chapter order. Chapters are then greedily concatenated into segments of up to 10 minutes, yielding long-form audio–text pairs aligned with the evaluation setting.

¹ 🤖 [google/translategemma-12b-it](https://huggingface.co/google/translategemma-12b-it)

² 🤖 [nvidia/parakeet-tdt-0.6b-v2](https://huggingface.co/nvidia/parakeet-tdt-0.6b-v2)

We further apply truecasing and punctuation restoration using NVIDIA NeMo’s (Kuchaiev et al., 2019) *PunctuationCapitalizationModel* (punctuation_en_bert), a BERT-based model (Devlin et al., 2019), converting lowercase, unpunctuated transcripts into well-formed text.

Collectively, these augmentations transform four short-form ASR corpora into long-form training data without requiring new recordings or manual annotation, demonstrating that existing resources can be effectively repurposed for long-form settings through concatenation and normalization alone.

2.3 Speech Translation (ST)

We use four datasets for the ST task:

EuroParl-ST. We apply the same long-form concatenation pipeline used for ASR to EuroParl-ST, generating long-form speech translations for the language pairs en-de and en-it.

CoVost. We apply a similar long-form concatenation pipeline to CoVost (Wang et al., 2020) to generate long-form speech translation pairs for en-zh.

NUTSHELL & LibriSpeech. We leverage respective transcripts and translate them to German, Italian, and Chinese.

2.4 Spoken Question Answering (SQA)

We rely on two SQA benchmarks and their translations:

LibriSQA. We use the open-form question subset of LibriSQA (Zhao et al., 2023), a spoken question-answering dataset built on LibriSpeech (Panayotov et al., 2015). We translate questions and answers into German, Italian, and Chinese as described earlier.

NUTSHELL. To address the lack of academic-domain coverage in LibriSQA, we adapt NUTSHELL for SQA. Using ASR transcripts, we prompt gemma-3-12b-it³ (Team et al., 2025) to generate five questions per transcript: four answerable from the content and one unanswerable.

2.5 Multiple Choice (MC)

To improve generalization and mitigate overfitting and catastrophic forgetting, as well as to prepare for

the surprise task, we incorporate multiple-choice style data.

LibriSQA (LibriMC). We use the multiple-choice subset of LibriSQA (Zhao et al., 2023) and translate questions into the missing three target languages.

MMSU. We leverage the Massive Multi-task Spoken Language Understanding and Reasoning Benchmark (MMSU) (Wang et al., 2025), using its multiple-choice questions to strengthen instruction-following capabilities.

2.6 SSUM

We use the following two datasets for SSUM, and their translations into the other target languages.

NUTSHELL. We use NUTSHELL (Züfle et al., 2025), which pairs ACL talk videos with corresponding paper abstracts. We treat abstracts as reference summaries.

YTSeg. We augment YTSeg with synthetic abstract-style summaries by filtering to academic domains (*Science, Technology, Education*) and prompting Qwen3.5-72B with transcript inputs and NUTSHELL-style examples, enforcing similar length and style.

2.7 Audio Chaptering (ACHAP)

Finally, for audio chaptering, we include two datasets, both of which are translated into all target languages:

YTSeg. For ACHAP, we utilize YTSeg and follow the same preprocessing as described in Section 2.2. We frame chaptering as a structured transcription task involving joint transcription, segmentation, and title generation, effectively extending the long-form ASR task. We use Markdown formatting for the structured output.

NUTSHELL. We augment NUTSHELL with chapter annotations generated by Qwen3-Omni⁴ using the ICL setup of Retkowski et al. (2026). We retain samples with 3–15 min duration, WER <30%, and 4–11 chapters.

2.8 General Instruction Following

To further enhance generalization and preserve instruction-following capabilities across domains, we incorporate general instruction-following data.

³ 🤖 google/gemma-3-12b-it

⁴ 🤖 Qwen/Qwen3-Omni-30B-A3B-Instruct

Dataset	# Samples	Initial (%)	p (% , T=2.0)
ASR	19,248	1.84	6.12
SQA	474,888	45.31	30.41
MC	380,056	36.26	27.21
SSUM	35,748	3.41	8.34
ST	29,343	2.80	7.56
ACHap	37,862	3.61	8.59
Instruct	71,013	6.78	11.76
Total	1,048,158	100.00	100.00

Table 1: Training data distribution with original proportions (init) and temperature-smoothed sampling probabilities ($T = 2.0$, where $p_i \propto n_i^{1/T}$).

TowerBlocks. We use TowerBlocks (Alves et al., 2024) to construct an augmented multi-modal corpus for instruction-following fine-tuning. TowerBlocks encompasses diverse subtasks including translation, named-entity recognition, post-editing, and paraphrase generation. We focus on the UltraChat subset, which provides a context-instruction format. We first filter samples matching this structure using Llama-3.1-8B⁵, tagging corresponding components. We then generate speech samples from the context using Kokoro-82M⁶, a state-of-the-art TTS model. This enables the model to refer to the speech during instruction-following fine-tuning.

3 Model Submissions

We submit two systems: an end-to-end model as our primary submission and a cascaded model as a contrastive system to analyze the trade-offs between the two approaches.

3.1 End-To-End Model

We select Qwen2.5-Omni⁷ (Xu et al., 2025a) as our primary end-to-end model and use LLamaFactory (Zheng et al., 2024) as the training framework. We chose this model for its strong multimodal instruction-following capabilities. Due to time and hardware constraints, we were unable to evaluate the more recent Qwen3-Omni (Xu et al., 2025c).

3.2 Cascaded Model

We additionally explore a cascaded pipeline that decomposes the task into ASR followed by text-based instruction following. For ASR, we use parakeet-tdt-0.6b-v2⁸, which gener-

ates transcripts from audio. These are passed to Qwen2.5-7B-Instruct⁹ together with task-specific prompts to produce final outputs.

To match the shared task format, we replace the <audio> field with a Transcript: prefix followed by the ASR transcript, enabling fully text-based processing. Unless stated otherwise, we use the same training data and hyperparameters as in the end-to-end setup. For MMSU, transcripts are generated with parakeet-tdt-0.6b-v2. Inference uses greedy decoding (beam size 1) with a maximum length of 4096 tokens.

4 Experimental Setup

We now detail our training recipe, covering prompt design, hyperparameters, data interleaving strategies, and evaluation criteria.

4.1 Training Strategy

To ensure our model strictly adheres to given instructions, we employ a fixed and restrictive system prompt throughout training and inference. For better instruction generalization, we create several different instruction prompts per task and per language, and each training and development instance is randomly assigned one of these prompts. The system prompt and instruction templates are provided in Tab. A1 in the Appendix.

We also experiment with Chain-of-Thought (CoT) style predictions by introducing special task tokens with noise initialization while keeping all other settings unchanged. The task tokens include <|asr|>, <|st|>, <|mc|>, <|sqa|>, <|achap|>, <|ssum|>, and <|instruct|>, along with language tokens <|en|>, <|de|>, <|it|>, and <|zh|>. For SQA, we additionally include answerability tokens <|answerable|> and <|unanswerable|>. The token order follows: task token, language token, answerability token (if applicable), and then the target output.

4.2 Data Preparation

Our training data exhibits significant class imbalance across tasks (see Tab. 1). To ensure adequate task visibility during training, we maintain datasets separately and apply interleaving strategies. We explore two approaches: (1) smoothed sampling, where we assign fixed probabilities per task, and (2) temperature-scaled sampling, where probabilities are derived from dataset sizes.

⁵ 🤖 meta-llama/Llama-3.1-8B

⁶ 🤖 hexgrad/Kokoro-82M

⁷ 🤖 Qwen/Qwen2.5-Omni-7B

⁸ 🤖 nvidia/parakeet-tdt-0.6b-v2

⁹ 🤖 Qwen/Qwen2.5-7B-Instruct

For smoothed sampling, we assign the following probabilities: ASR (10%), SQA (31%), MC (9%), SSUM (14%), ST (13%), ACHAP (14%), and general instruction following (9%). We deliberately reduced the MC probability since it was only a potential surprise task and we wanted to avoid overemphasizing it during training.

For temperature-scaled sampling, let n_i denote the number of training instances in dataset i , $N = \sum_i n_i$ the total, and $f_i = n_i/N$ the original size-proportional fraction. We apply temperature scaling to obtain interleaving probabilities:

$$p_i = \frac{n_i^{1/T}}{\sum_j n_j^{1/T}}, \quad (1)$$

with $T > 0$ interpolating between size-proportional sampling ($T = 1$) and uniform sampling ($T \rightarrow \infty$). We adopt $T = 2$, following prior work showing improvements on the ST task (Li et al., 2025), which corresponds to sampling proportional to the square root of dataset size ($p_i \propto \sqrt{n_i}$).

4.3 Hyperparameters

We experimented with various training configurations and found that an effective batch size of 4 with a learning rate of 1.0e-4 yielded the best results, outperforming configurations with larger batch sizes and learning rates ranging from 1.0e-5 to 2.0e-4. To prevent out-of-memory errors, we set the token cutoff length to 28,000 tokens. Since 15 minutes of audio requires approximately 22,500 tokens, this provides sufficient headroom for model predictions. We set the warmup ratio to 0.1 with a total of 60,000 update steps, and apply LoRA (Pham et al., 2021; Hu et al., 2022) with rank 32.

4.4 Evaluation

For model selection, we evaluate on the MCIF (Multimodal Crosslingual Instruction-Following) benchmark (Papi et al., 2026), YTSeg, and LibriMC. We additionally include an unused YTSeg validation subset with ACHAP translations for German, Italian, and Chinese. During training, we extract a small subset from each development split to monitor evaluation loss and inform checkpoint selection. We apply the following evaluation protocols per task:

- **ASR, ST, SQA, SSUM:** We follow the automated evaluation protocols of Papi et al. (2026).

- **ACHAP:** We follow Retkowski et al. (2026), employing Collar-F1 (with ± 3 second tolerance) to assess segmentation quality and the Global Concatenation protocol with BERTScore to measure title quality, as implemented in the chunkseg package¹⁰.

- **MC:** We compute accuracy as $\frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i]$, where predictions are normalized by stripping whitespace and must exactly match one of the choices {A, B, C, D}.

- **Surprise Task:** As the nature of the surprise task is unknown at training time, we train the model to identify the most similar known task and apply the corresponding behavior.

4.5 Results

We report results on the MCIF long-form track as well as on the ACHAP and MC tasks, which are evaluated separately due to their distinct evaluation protocols.

4.5.1 MCIF Results

Tab. 2 presents results on the MCIF long-form track with fixed and mixed prompts and beam size 1. We compare two baselines: Qwen2.5-Omni (end-to-end) and Cascaded (Section 3.2).

Baselines. The cascaded baseline demonstrates the advantage of using an ASR model specifically optimized for long-form audio, achieving 5.88% WER and 80.81% COMET score on translation. In contrast, the end-to-end baseline outperforms the cascaded approach on SQA and SSUM, but its inferior ASR and ST performance (53.40% WER, 68.65% COMET) reveals that it was not specifically designed for long-form audio.

Fine-tuning. Fine-tuning substantially improves both systems. We observe faster convergence and better final results with temperature-scaled data sampling (row 4) compared to smoothed sampling (row 3), and use it as our base for further optimization.

This confirms that $T=2$ consistently is more stable heuristic than smoothed sampling across all tasks (see Table 2, rows 3 vs. 4), extending the finding of Li et al. (2025) from text-only multi-task learning to the multimodal speech instruction-following setting. This suggests that square-root

¹⁰<https://github.com/retkowski/chunkseg>

Models	SQA ($\uparrow$)		SSUM ($\uparrow$)		ASR ($\downarrow$)		ST ($\uparrow$)	
	Fix.	Mix.	Fix.	Mix.	Fix.	Mix.	Fix.	Mix.
Baselines								
(1) Qwen 2.5 Omni	30.78	32.94	14.21	17.87	53.40	35.35	68.65	70.79
(2) Cascaded	27.48	27.66	13.31	13.40	<u>5.88</u>	6.85	<u>80.81</u>	80.46
Fine-tuned								
(3) Smoothed Sampling	36.16	36.42	28.77	28.76	30.59	33.61	75.58	74.90
(4) Temp. Sampling ($T = 2$)	37.75	37.87	28.87	28.99	27.58	38.43	76.10	75.96
(5) N2	39.98	39.98	29.05	28.75	25.98	<i>30.46</i>	74.98	74.94
(6) N2+Avg	40.68	40.86	26.06	29.33	29.36	34.48	73.35	75.42
(7) N2+IT+Avg	<u>40.42</u>	<u>40.31</u>	<i>26.10</i>	<i>29.41</i>	37.65	36.88	<i>73.94</i>	<i>76.01</i>
(8) N2+IT+SHAS	-11.09	-12.14	2.93	5.90	<u>11.72</u>	<u>11.25</u>	<u>80.62</u>	<u>80.84</u>
(9) Chain-of-Thought (CoT)	34.88	35.50	26.56	29.74	79.24	80.04	42.09	43.69
(10) Cascaded+FT	33.36	32.09	<u>28.55</u>	27.33	5.90	<u>9.76</u>	83.72	83.75

Legend: (5) continued fine-tuning on in-domain (NUTSHELL) data; (6) checkpoint averaging of (5); (7) additional Italian in-domain fine-tuning with averaging; (10) cascaded system with fine-tuning.

Table 2: MCIF long track comparison of fixed vs. mixed prompt evaluation (macro-averaged). Best results are in **bold**, second-best are underlined and third-best in *italic*. Shaded rows indicate submitted models.

proportional sampling is a robust default for multi-task speech training regardless of modality.

In-Domain Adaptation and Checkpoint Averaging. Continuing training on NUTSHELL, an in-domain dataset of scientific talks similar to the evaluation data, yields clear improvements (row 5), supporting our hypothesis that in-domain adaptation benefits long-form performance. Checkpoint averaging of the best two checkpoints (row 6) provides further gains, whereas averaging additional checkpoints degraded performance. Since performance on Italian remained suboptimal (see Table A9), we additionally fine-tuned row (4) with Italian in-domain data and averaged the best checkpoint with those from row (6), yielding row (7). This model shows modest improvements on SSUM and ST tasks.

Segmented Evaluation. To diagnose whether poor ASR results stem from task confusion rather than fundamental ASR limitations, we applied segmented evaluation (SHAS) to row (7). The degradation on SQA and SSUM is expected, but acceptable ASR performance suggests that long-form nature, rather than the task itself, drives the poor unsegmented results.

Chain-of-Thought Conditioning. We evaluate CoT-style task-token conditioning (row 9) (Koneru et al., 2025), but observe degraded performance, especially for ASR (79.24% WER).

Task prediction collapses: ASR inputs are almost always misclassified as SSUM, despite only

a slight data imbalance (~2%). Explicit ASR predictions are rare (3 fixed, 2 mixed) and occur only when the ground truth is SQA, indicating a failure to learn task discrimination.

We hypothesize that the model defaults to SSUM because it is both slightly more frequent and structurally closer to ASR (i.e., grounded in transcript-level content), making it an easier fallback. This suggests that prefix-based task routing remains weakly grounded and prone to shortcut behavior, requiring stronger balancing or supervision to be effective.

Cascaded System. Fine-tuning the cascaded system substantially improves performance on SQA (27.48 $\rightarrow$ 33.36%) and SSUM (13.31 $\rightarrow$ 28.55%) while preserving strong ASR performance (5.88 $\rightarrow$ 5.90% WER). The cascaded model achieves the best overall ST result (83.72% COMET) among all systems. These results highlight that the cascaded approach benefits from high-quality intermediate transcripts and excels at transcription-sensitive tasks. However, performance on semantic tasks such as SQA and SSUM lags the end-to-end model, likely due to error propagation from ASR transcripts and loss of audio-specific information. Additionally, training data aligned with the end-to-end setup may not be optimal for text-based instruction following, potentially limiting the cascaded model’s ability to handle semantic reasoning tasks.

Model Selection. Model selection was based on robustness across all tasks. Row (7) consistently

ranks as the second or third best performer across metrics, closely following the top model. For the surprise task, we hypothesized that the Italian fine-tuning checkpoint, when averaged with others, could mitigate catastrophic forgetting and improve generalization (Ugan et al., 2025). Thus, we selected row (7) as our primary submission and row (10) as a contrastive system. Although the cascaded+FT model underperforms on SQA and SSUM, its superior ASR and ST performance and qualitative analysis suggest it may handle challenging surprise task instances.

4.5.2 ACHAP and MC Results

Results on ACHAP and MC tasks are presented in Table A11 in the Appendix. We evaluated MC only on English, as we identified it as a potential surprise task and reserved other languages for evaluation robustness. For ACHAP, we used 50 unused samples from the YTSeg development split, translating them into German, Italian, and Chinese for multilingual evaluation.

4.6 Re-ranking

To improve generation quality beyond single-pass decoding, we generate $N = 17$ candidate outputs per segment: one greedy decode, one greedy decode with SHAS-based segmentation (Huber et al., 2023; Tsiamas et al., 2022), and 15 sampled candidates. We then apply re-ranking to select the best hypothesis.

A key challenge is that the task identity is unknown at inference time: the re-ranker therefore receives audio, the original prompt instructions, and candidates. This realistic constraint prevents task-specific strategies and requires a single method to generalize across several tasks.

Reranking Strategies. We evaluate six reranking strategies: *Likelihood* scores each candidate independently via the model’s conditional probability; *Comparison* presents all candidates simultaneously and asks the model to select the best; *Pairwise* runs a sequential tournament where the greedy candidate serves as champion and faces each remaining candidate in turn; *Pairwise Round-Robin* compares every pair of candidates once and selects the candidate with the most wins; *Pairwise Bracket* organises comparisons as a single-elimination bracket, mitigating positional bias from sequential tournaments; and *MBR* selects the candidate with the highest average chrF similarity to all

others, requiring no model inference. Additional details, including inference costs, can be found in Tab. A2.

Method	ASR (↓)	SQA (↑)	SSUM (↑)	ST (↑)	Impr.
Greedy	40.77	37.24	29.38	75.28	—
Oracle	-32.10	+14.42	+3.85	+6.11	+14.12
Likelihood	-24.93	-11.06	-8.60	+0.93	+1.55
Comparison	+4.16	-4.25	-2.03	-1.35	-2.95
Pairw.	-5.62	-3.78	-1.71	-1.02	-0.22
Pairw. RR	-1.39	-3.36	-1.84	-1.51	-1.33
Pairw. Brack.	-1.83	-3.25	-1.52	-0.63	-0.89
MBR	+3.09	+0.22	-0.79	+0.27	-0.85
Lik. + MBR	-19.28	-3.33	-2.19	+1.09	+3.71
Casc. Pairw.	+3.39	-3.32	-2.29	-1.83	-2.71
Casc. Pairw. RR	+14.77	-3.66	-1.45	-0.65	-5.13
Casc. Pairw. Brack.	+12.43	-2.97	-1.76	-1.24	-4.60

Table 3: Re-ranking results. Values show Δ vs. greedy. ASR uses WER (lower $\downarrow$ is better); SQA/SSUM/ST use BERTScore/COMET (higher $\uparrow$ is better). Casc. = Whisper + Gemma two-stage pipeline; Pairw. = Pairwise; RR = Round-Robin; Lik. + MBR = Likelihood and MBR combined with a tiebreaking pairwise comparison. Greedy row shows absolute scores. Impr. = mean signed improvement (ASR sign-flipped).

Reranking Models. We primarily use Qwen2.5-Omni¹¹ (Xu et al., 2025a) as the re-ranking model, since it can directly condition on the audio input. We additionally experiment with a two-stage pipeline combining whisper-large-v3¹² (Radford et al., 2022) for audio transcription and gemma-3-12b-it¹³ (Team et al., 2025) for text-based candidate comparison, applying all three pairwise strategies.

4.6.1 Reranking Results

Results averaged across languages are shown in Tab. 3. The oracle upper bound reveals substantial potential gains for ASR (-32.1 WER points) and SQA ($+14.4$), indicating high diversity among the 17 candidates. For SSUM and ST, the oracle improvements are considerably smaller ($+3.9$ and $+2.0$ respectively), suggesting less variance and thus limited headroom for re-ranking.

In practice, no single method reliably captures these gains across all tasks. *Likelihood* achieves the strongest ASR improvement (-24.9 WER points), but at the cost of large degradations on SQA and SSUM. *MBR* is conservative: it preserves SQA quality but provides no ASR benefit. Combining both signals via *Lik. + MBR*, which resolves disagreements between Likelihood and MBR through

¹¹ 🤖 Qwen/Qwen2.5-Omni-7B

¹² 🤖 openai/whisper-large-v3

¹³ 🤖 google/gemma-3-12b-it

a single pairwise comparison, retains most of the ASR gain (−19.3) while substantially reducing the SQA degradation (−2.9), yielding the best overall score. Per-language results (Tabs. A3 to A6 Appendix) show consistent gains only for English and Chinese, while German and Italian show no reliable improvement over greedy. For the final submission, we therefore apply *Lik.* + *MBR* re-ranking only for English and Chinese.

4.6.2 Reranking Analysis

We analyze two aspects of re-ranker behavior that explain the inconsistent results: positional bias and spurious SHAS-candidate selection.

Positional Bias. Pairwise methods are susceptible to spurious preferences for whichever candidate appears first (A-bias) or second (B-bias) in a comparison. We quantify this as

$$P(\text{pick A} \mid \text{B better}) - P(\text{pick B} \mid \text{A better}) \quad (2)$$

where positive values indicate A-bias and negative values B-bias. Results are shown in Tab. A7 in App. C. Sequential *Pairwise* exhibits strong A-bias (+35.3,pp on average, up to +56.8,pp for German), which is expected since the greedy candidate always starts as champion and thus always occupies the first position. *Round-Robin* nearly eliminates this bias (+3.2,pp), since every pair is compared once in each order. *Bracket* introduces a slight B-bias (−5.5,pp) with high variance across languages. These systematic biases likely explain the inconsistent performance of sequential and bracket methods in Tab. 3.

SHAS candidate selection. A second source of error is the spurious selection of the SHAS-segmented candidate for tasks where accurate segmentation provides no benefit. As shown in Tab. A8 in App. C, *Likelihood* selects SHAS frequently for SQA (22.7%) and SSUM (34.5%), directly explaining its large degradations. *MBR* avoids this entirely, the SHAS output tends to be less fluent and therefore scores poorly under chrF-based similarity, but consequently fails to select it for ASR and ST, where it would be beneficial. *Lik.*, +, *MBR* strikes the best balance: spurious SHAS selection drops sharply for SQA (7.4%) and SSUM (5.5%), while ASR (19.0%) and ST (25.4%) retain it at useful rates. This explains why *Lik.*, +, *MBR* achieves the best overall score in Tab. 3.

Method	EN			ZH		
	SQA↑	SSUM↑	ASR↓	SQA↑	SSUM↑	ST↑
(7)	44.91	23.44	37.65	39.31	39.57	79.06
(7)+Reranker	44.30	23.10	21.39	40.52	40.11	80.61

Table 4: MCIF-long fixed-prompt evaluation comparing N2+IT+Avg and its reranked variant. Best values per column are in **bold**.

These results suggest a general principle for multi-task re-ranking without task identity: likelihood alone is too aggressive, MBR alone is too conservative, and their combination provides the best tradeoff by using MBR as a regularizer that suppresses spurious likelihood-driven candidate selection. We recommend *Lik.*+*MBR* as a default re-ranking strategy for multi-task speech instruction-following systems where task identity is unavailable at inference time.

4.7 Submission setting

We selected model (7) as our main submission setup and additionally apply the re-ranking strategy as explained in Section 4.6. The re-ranking showed decreasing performance on Italian and German targets, we report only the English and Chinese results on the MCIF long track in Tab. 4. While English shows slight degradation on SQA and SSUM, the ASR improvement is substantial (37.65 → 21.39 WER). Chinese shows consistent improvements across all tasks. These results motivate us to submit the re-ranking strategy for the English and Chinese tracks.

5 Conclusion

This work presents five contributions to long-form speech instruction following: (1) a general data augmentation framework for constructing long-form training data from short-form sources; (2) empirical evidence supporting temperature-scaled sampling in multimodal settings; (3) a negative result on CoT task-token conditioning, revealing a collapse in task discrimination due to weak grounding and task similarity rather than simple data imbalance; (4) a systematic analysis of re-ranking under the realistic constraint of unknown task identity; and (5) a combined *Lik.*+*MBR* reranking strategy that mitigates the trade-off between transcription and semantic tasks.

We release our data augmentation code and model checkpoints to support future research on long-form multimodal instruction following.

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A Appendix

B Data

Prompts & Instructions We employ an LLM to generate paraphrased variants of manually designed instruction prompts. For readability, we report a representative subset in Table [A1](#).

System Prompts			
(1) You are a helpful assistant specialized in audio and video processing tasks. You follow instructions exactly and don't provide additional explanations or follow-up questions.			
(2) You are a helpful assistant specialized in audio processing tasks. You follow instructions exactly and don't provide additional explanations or follow-up questions.			
Task	Lang.	#Instr.	Representative instruction variants
ASR	EN	30	(1) Transcribe the entire audio recording into plain text. (2) Generate a faithful transcript of this audio segment. (3) Convert this audio into text exactly as spoken.
	DE	10	(1) Konvertieren Sie dieses Audio in deutschen Text. (2) Anhören und ins Deutsche übersetzen. (3) Was wird gesagt? Geben Sie die deutsche Übersetzung an.
	IT	10	(1) Converti questo audio in testo italiano. (2) Ascolta e traduci in italiano. (3) Cosa viene detto? Fornisci la traduzione in italiano.
ST	ZH	10	(1) 请将其翻译成中文。 (2) 请提供这段录音的中文翻译。 (3) 翻译此音频为中文。
	EN	10	(1) Organize this recording into meaningful chapters. For each one, include a Markdown heading (# Title) enclosed by two newlines (\n\n), followed by its transcript. (2) Break this audio into coherent sections as chapters. For every chapter, write a Markdown heading (# Title) with two newlines around it (\n\n), then add the transcript for that part. (3) Create coherent chapter segments for this audio. Each chapter should have a Markdown heading (# Title) with two surrounding newlines (\n\n), then the transcript of that chapter.
	DE	10	(1) Unterteile dieses Audio in zusammenhängende Kapitel und transkribiere jedes davon. Beginne jedes Kapitel mit einer Markdown-Überschrift (# Title), umgeben von zwei Zeilenumbrüchen (\n\n), und füge dann das Transkript an. (2) Segmentiere dieses Audio in zusammenhängende Kapitel. Für jedes Kapitel gib eine Markdown-Überschrift (# Title) mit zwei umgebenden Zeilenumbrüchen (\n\n) aus und füge danach das Transkript dieses Abschnitts an. (3) Teile diese Aufnahme in sinnvolle Kapitel auf. Für jedes Kapitel schreibe eine Markdown-Überschrift (# Title), die von zwei Zeilenumbrüchen (\n\n) umgeben ist, und danach den Kapiteltext.
ACHAP	IT	10	(1) Dividi questo audio in capitoli logici. Ogni capitolo deve iniziare con un'intestazione Markdown (# Title) circondata da due nuove righe (\n\n), seguita dal testo trascritto. (2) Suddividi questa registrazione in capitoli coerenti. Per ogni capitolo, inserisci un'intestazione Markdown (# Title) con due nuove righe attorno (\n\n), poi aggiungi la trascrizione del capitolo. (3) Segmenta questa registrazione seguendo i confini dei capitoli. Per ogni capitolo, usa un'intestazione Markdown (# Title) avvolta da due nuove righe (\n\n), poi fornisci la trascrizione.
	ZH	10	(1) 请将这段音频按逻辑分成多个章节。每章以 Markdown 标题 (# Title) 开头，标题前后各有两个换行 (\n\n)，并在后面附上对应转写文本。 (2) 请按章节边界对这段录音进行切分。每个章节使用 Markdown 标题 (# Title)，标题前后各两个换行 (\n\n)，然后提供该章节转写。 (3) 请把这段音频章节化为连贯部分。每个部分都输出 Markdown 标题 (# Title)，标题前后各两个换行 (\n\n)，然后写该部分转写。
	EN	10	(1) Reply with only the choice. (2) Reply with "The answer is" followed by the choice. (3) Reply with "Option" followed by the choice.
MC	DE	10	(1) Antworte nur mit der Auswahl. (2) Antworte mit „Die Antwort ist“ gefolgt von der Auswahl. (3) Antworte mit „Option“ gefolgt von der Auswahl.
	IT	10	(1) Rispondi solo con la scelta. (2) Rispondi con "La risposta è" seguito dalla scelta. (3) Rispondi con "Opzione" seguito dalla scelta.
	ZH	10	(1) 只回答选项字母。 (2) 请回答“答案是”加上选项字母。 (3) 请回答“选项”加上选项字母。
SQA	EN	10	(1) Based on the English content, respond to this question with a brief answer: (2) Use the English content to provide a concise answer to the question below: (3) Refer to the English content to answer the question. Be concise:
	DE	20	(1) Beantworte die Frage basierend auf dem Inhalt kurz: (2) Nutze den Inhalt und antworte prägnant auf die folgende Frage: (3) Beantworte die Frage basierend auf dem englischen Inhalt kurz:
	IT	20	(1) In base al contenuto, rispondi brevemente alla domanda: (2) Usa il contenuto per fornire una risposta concisa: (3) In base al contenuto inglese, rispondi brevemente alla domanda:
SSUM	ZH	20	(1) 请基于内容，简要回答这个问题: (2) 根据内容，对下列问题给出简洁回答: (3) 请基于英文内容，简要回答这个问题:
	EN	10	(1) Summarize the English audio. (2) Provide a concise summary of the English audio. (3) Summarize the English audio in at most {x} words.
	DE	10	(1) Fassen Sie das englische Audio zusammen. (2) Erstellen Sie eine knappe Zusammenfassung des englischen Audios. (3) Fassen Sie das englische Audio in maximal {x} Wörtern zusammen.
SSUM	IT	10	(1) Riassumi l'audio inglese. (2) Fornisci un riassunto conciso dell'audio in inglese. (3) Riassumi l'audio inglese in massimo {x} parole.
	ZH	10	(1) 概括英文音频内容。 (2) 请简要总结英文音频的主要内容。 (3) 请将英文音频内容概括为不超过{x}个词。

Table A1: Compact summary of the instruction templates used in training after deduplication. We report the number of unique templates per task/language and show three representative examples for each setting. For NUTSHELL data we used System Prompt (1) as and the others (2). In SSUM, the {x} was replaced with the number of words in the target summary.

C Reranking Strategies

Setting. We generate $N = 17$ candidates per segment: one greedy decode, one greedy decode with SHAS-based segmentation (Tsiamas et al., 2022), and 15 samples (temperature 0.8). Candidates are produced by our fine-tuned model, while re-ranking is performed by the base Qwen2.5-Omni-7B (Xu et al., 2025a), which can directly condition on the audio input. For the cascaded methods, audio is first transcribed with whisper-large-v3 (Radford et al., 2022) and the resulting transcript is passed together with all candidates to gemma-3-12b-it (Team et al., 2025) for text-based comparison. An overview of the re-ranking strategies and the number of model calls required for each is given in Tab. A2.

Method	Description	Model calls
Likelihood	Each candidate is scored independently by computing $\log p(\text{candidate} \mid \text{audio})$ with an external model. The candidate with the highest likelihood is selected.	N
Comparison	The model receives the audio, the task, and all candidates simultaneously and is prompted to identify the best candidate.	1
Pairwise	A sequential tournament: candidate 0 (greedy) acts as champion and is compared against candidates 1 through $N-1$ in order; the winner of each comparison advances.	$N-1$
Pairwise Round-Robin	Every pair of candidates is compared once; the candidate with the most wins is selected. Unbiased with respect to candidate order.	$\frac{N(N-1)}{2}$
Pairwise Bracket	A single-elimination bracket where winners are matched against winners across $\log_2 N$ rounds, avoiding the positional bias of the sequential tournament.	$N-1$
MBR	Minimum Bayes Risk (Kumar and Byrne, 2004): the candidate with the highest average chrF similarity to all other candidates is selected.	0

Table A2: Overview of re-ranking methods. For model-based methods, the original instruction is included in the prompt of the reranker since the task is determined at inference time.

Per-language results. We use the MCIF test set (Papi et al., 2026) to evaluate our re-ranking methods using the official MCIF evaluation code. Results per language are shown in Tab. A3 (en), Tab. A4 (de), Tab. A5 (it), and Tab. A6 (zh). Re-ranking yields consistent improvements only for English, where Likelihood achieves the largest gain, driven primarily by a strong improvement on ASR. For Chinese, modest improvements are observed with Likelihood and MBR. For German and Italian, no method reliably improves over greedy across all tasks.

Analysis. Tab. A7 reports positional bias for each pairwise re-ranking method and language, measured as

$$P(\text{pick A} \mid \text{B better}) - P(\text{pick B} \mid \text{A better}). \quad (3)$$

Positive values indicate a spurious preference for the first candidate, negative values for the second. Tab. A8 reports the proportion of segments for which each re-ranking method selects the SHAS-segmented candidate, broken down by task. Both tables complement the analysis in Section 4.6.

D Speech Translation

We perform English to German, Italian, and Chinese speech translation using `translategemma-12b-it` (Finkelstein et al., 2026). Input speech is first transcribed into English text and embedded within a structured conversational format, from which the relevant response segments are extracted for translation. To efficiently handle large-scale datasets, we employ a streaming JSON parser (`ijson`), enabling memory-efficient sequential processing.

Inference is conducted in batches of size 8, with the model loaded in `bfloat16` precision. Inputs are formatted using the model-specific chat template and tokenized with dynamic padding before being transferred to the appropriate device. Decoding is performed using deterministic (greedy) generation with a maximum output length depending on the dataset. To improve generation stability and mitigate repetition artifacts, we apply a repetition penalty around 1.1 and enforce a 5-gram repetition constraint.

Method	ASR ↓	SQA ↑	SSUM ↑	Impr.
Greedy	40.77	38.43	23.43	—
Oracle	-32.10	+14.84	+5.51	+17.48
Likelihood	-24.93	-9.21	-2.52	+4.40
Comparison	+4.16	-4.63	-2.10	-3.63
Pairw.	-5.62	-3.19	-1.94	+0.16
Pairw. RR	-1.39	-3.37	-2.07	-1.35
Pairw. Brack.	-1.83	-2.67	-1.32	-0.72
MBR	+3.09	+1.59	-0.41	-0.64
Lik. + MBR	-19.28	-2.94	-2.18	+4.72
Casc. Pairw.	+3.39	-2.46	-2.28	-2.71
Casc. Pairw. RR	+14.77	-3.74	-1.50	-6.67
Casc. Pairw. Brack.	+12.43	-1.98	-1.46	-5.29

Table A3: Re-ranking results — English. Values show Δ vs. greedy. ASR uses WER (lower ↓ is better); QA/SUM/ST use BERTScore/COMET (higher ↑ is better). Casc. = Whisper + Gemma two-stage pipeline; Pairw. = Pairwise; RR = Round-Robin; Lik. + MBR = Likelihood and MBR combined with a tiebreaking pairwise comparison. Greedy row shows absolute scores. Impr. = mean signed improvement (ASR sign-flipped). **Bold**: best method per column.

Method	SQA ↑	SSUM ↑	ST ↑	Impr.
Greedy	36.77	25.60	72.91	—
Oracle	+15.76	+4.06	+5.49	+8.44
Likelihood	-14.29	-8.01	+2.77	-6.51
Comparison	-2.55	-2.55	-1.47	-2.19
Pairw.	-2.68	-1.09	-1.22	-1.66
Pairw. RR	-2.26	-1.22	-1.05	-1.51
Pairw. Brack.	-1.79	-0.86	-1.25	-1.30
MBR	+0.70	-1.03	-0.46	-0.26
Casc. Pairw.	-1.98	-2.05	-1.46	-1.83
Casc. Pairw. RR	-2.98	-1.42	-0.91	-1.77
Casc. Pairw. Brack.	-1.69	-1.74	-1.51	-1.65

Table A4: Re-ranking results — German. Values show Δ vs. greedy. ASR uses WER (lower ↓ is better); QA/SUM/ST use BERTScore/COMET (higher ↑ is better). Casc. = Whisper + Gemma two-stage pipeline; Pairw. = Pairwise; RR = Round-Robin. Greedy row shows absolute scores. Impr. = mean signed improvement (ASR sign-flipped). **Bold**: best method per column.

Method	SQA ↑	SSUM ↑	ST ↑	Impr.
Greedy	35.60	28.38	74.80	—
Oracle	+13.76	+2.72	+6.27	+7.58
Likelihood	-16.13	-8.78	+1.19	-7.91
Comparison	-5.05	-2.26	-3.06	-3.46
Pairw.	-4.32	-2.53	-2.22	-3.02
Pairw. RR	-4.46	-2.23	-1.96	-2.88
Pairw. Brack.	-4.99	-2.66	-1.72	-3.12
MBR	-1.49	-1.74	-0.33	-1.19
Lik. + MBR	-5.03	-3.21	+0.43	-2.60
Casc. Pairw.	-2.80	-3.18	-4.70	-3.56
Casc. Pairw. RR	-3.22	-2.39	-1.80	-2.47
Casc. Pairw. Brack.	-2.51	-2.52	-3.56	-2.86

Table A5: Re-ranking results — Italian. Values show Δ vs. greedy. ASR uses WER (lower ↓ is better); QA/SUM/ST use BERTScore/COMET (higher ↑ is better). Casc. = Whisper + Gemma two-stage pipeline; Pairw. = Pairwise; RR = Round-Robin; Lik. + MBR = Likelihood and MBR combined with a tiebreaking pairwise comparison. Greedy row shows absolute scores. Impr. = mean signed improvement (ASR sign-flipped). **Bold**: best method per column.

Method	SQA $\uparrow$	SSUM $\uparrow$	ST $\uparrow$	Impr.
Greedy	38.17	40.09	78.12	—
Oracle	+13.32	+3.11	+6.56	+7.66
Likelihood	-4.61	-15.10	-1.16	-6.96
Comparison	-4.78	-1.20	+0.48	-1.83
Pairw.	-4.94	-1.28	+0.38	-1.95
Pairw. RR	—	—	—	—
Pairw. Brack.	-3.55	-1.24	+1.09	-1.23
MBR	+0.09	+0.01	+1.61	+0.57
Lik. + MBR	+0.04	-0.43	+1.65	+0.42
Casc. Pairw.	-6.03	-1.65	+0.68	-2.33
Casc. Pairw. RR	-4.68	-0.47	+0.77	-1.46
Casc. Pairw. Brack.	-5.69	-1.31	+1.35	-1.88

Table A6: Re-ranking results — Chinese. Values show Δ vs. greedy. ASR uses WER (lower $\downarrow$ is better); QA/SUM/ST use BERTScore/COMET (higher $\uparrow$ is better). Casc. = Whisper + Gemma two-stage pipeline; Pairw. = Pairwise; RR = Round-Robin; Lik. + MBR = Likelihood and MBR combined with a tiebreaking pairwise comparison. Greedy row shows absolute scores. Impr. = mean signed improvement (ASR sign-flipped). **Bold**: best method per column.

Method	en	de	it	zh	Avg. Bias
Comparison	8.5 / 16 (last)	7.4 / 16 (first)	7.7 / 16 (first)	7.6 / 16 (first)	7.8 / 16 (first)
Pairw.	+30.0 (A)	+56.8 (A)	+49.3 (A)	+5.1 (A)	+35.3 (A)
Pairw. RR	+2.3 (A)	+4.0 (A)	+3.4 (A)	—	+3.2 (A)
Pairw. Brack.	-8.6 (B)	+11.3 (A)	+1.9 (A)	-26.8 (B)	-5.5 (B)
Casc. Pairw.	+19.0 (A)	+20.8 (A)	+17.8 (A)	+10.6 (A)	+17.1 (A)
Casc. Pairw. RR	+1.1 (A)	+3.6 (A)	+2.5 (A)	+2.1 (A)	+2.3 (A)
Casc. Pairw. Brack.	-21.2 (B)	-21.5 (B)	-24.9 (B)	-29.1 (B)	-24.2 (B)

Table A7: Positional preference of re-ranking methods. For pairwise methods: position bias = $P(\text{pick A} \mid \text{B oracle better}) - P(\text{pick B} \mid \text{A oracle better})$, with preferred position in parentheses (A = first, B = second); values near 0 indicate no bias. For *Comparison*: average selected candidate index (0 = greedy, 16 = SHAS), where 8 would be unbiased. Casc. = Whisper + Gemma cascade; Pairw. = Pairwise; RR = Round-Robin.

Method	ASR	SQA	SSUM	ST
Likelihood	19.0	22.7	34.5	36.5
MBR	0.0	0.0	0.0	0.0
Lik. + MBR	19.0	7.4	5.5	25.4

Table A8: Rate at which the SHAS-segmented candidate is selected per task (%). Ideally the re-ranker should prefer SHAS for ASR and ST (where accurate segmentation helps) but not for QA and SUM.

E Results

Per language evaluations While our main results (Section 4.5) present language-averaged scores, Tables A9 and A10 provide per-language breakdowns for all four languages on the MCIF long-form track with fixed and mixed prompts respectively. Table A11 shows MC results (English only) and AChap results (all languages). These analyses reveal substantial performance variation across languages, with some configurations (e.g., row 7) generalizing robustly while others (e.g., row 5) excel on specific languages. This granular view helps identify how language-specific challenges—morphological complexity, data scarcity, and pre-training bias—affect different tasks and languages.

ID	Method	EN			ZH			DE			IT		
		SQA	SSUM	ASR	SQA	SSUM	ST	SQA	SSUM	ST	SQA	SSUM	ST
Baselines													
(1)	Omni	24.93	19.34	53.40	43.03	9.59	75.88	28.34	11.55	61.16	26.82	16.35	68.92
(2)	Cascaded	25.07	19.36	5.88	26.68	10.31	84.66	31.64	11.15	76.55	26.53	12.41	81.23
Fine-tuned													
(3)	Temp	37.75	23.03	30.59	36.83	39.43	78.24	35.47	24.86	73.42	34.58	27.75	75.08
(4)	T=2	41.46	22.10	27.58	38.77	39.64	80.20	35.22	25.58	73.35	35.53	28.14	74.75
(5)	N2	44.41	23.95	25.98	41.51	39.44	78.20	37.63	25.47	72.33	36.36	27.32	74.40
(6)	N2+Avg	45.00	23.85	29.36	39.82	26.23	72.42	39.82	26.23	72.42	38.09	27.93	75.21
(7)	N2+IT+Avg	44.91	23.44	37.65	39.64	26.30	73.10	39.64	26.30	73.10	37.49	28.36	75.62
(8)	N2+IT+Avg+SHAS	-19.48	0.87	11.72	2.63	-2.78	84.32	-10.50	5.25	77.83	-16.99	8.38	79.72
(9)	CoT	34.49	24.29	79.24	35.06	26.70	39.97	35.06	26.70	39.97	34.92	28.53	46.33
(10)	Cascaded+FT	33.02	22.02	5.90	38.50	39.82	85.00	31.57	23.74	81.75	30.36	28.62	84.42

Legend: (3) temperature-based sampling; (4) temperature-based sampling with $T = 2$; (5) continued fine-tuning on in-domain NUTSHELL data after (4); (6) checkpoint averaging of (5); (7) continued fine-tuning on Italian NUTSHELL data with checkpoint averaging; (8) (7) with SHAS; (9) chain-of-thought prompting; (10) cascaded system with fine-tuning.

Table A9: MCIF-long fixed-prompt evaluation results across tasks (SQA, SSUM, ASR, ST) and languages.

ID	Method	EN			ZH			DE			IT		
		SQA	SSUM	ASR	SQA	SSUM	ST	SQA	SSUM	ST	SQA	SSUM	ST
Baselines													
(1)	Omni	33.15	17.57	35.35	39.59	21.25	75.74	30.37	16.04	66.61	28.66	16.63	70.01
(2)	Cascaded	26.97	18.12	6.85	28.87	12.51	84.40	28.06	11.08	76.70	26.72	11.88	80.28
Fine-tuned													
(3)	Temp	39.37	22.67	33.61	36.60	39.81	78.72	35.35	24.75	72.84	34.34	27.79	73.15
(4)	T=2	42.30	22.18	38.43	38.38	39.51	79.60	34.87	26.16	73.45	35.92	28.10	74.82
(5)	N2	44.46	23.35	30.46	41.34	39.26	78.35	38.20	25.12	72.92	35.93	27.25	73.56
(6)	N2+Avg	45.15	24.27	34.48	40.80	39.37	79.60	40.25	26.07	72.71	37.23	27.62	73.96
(7)	N2+IT+Avg	45.15	23.75	36.88	40.70	39.50	79.11	38.33	26.13	73.39	37.04	28.25	75.54
(8)	N2+IT+SHAS	-17.74	0.74	11.25	-0.95	7.48	84.25	-13.47	6.89	77.98	-16.39	8.47	80.30
(9)	CoT	37.29	24.06	80.04	35.30	39.88	44.82	34.74	26.56	41.03	34.66	28.44	45.22
(10)	Cascaded FT	32.49	22.06	9.76	37.62	38.03	85.19	29.40	21.68	81.58	28.83	27.56	84.47

Legend: (3) temperature-based sampling; (4) temperature-based sampling with $T = 2$; (5) continued fine-tuning on in-domain NUTSHELL data after (4); (6) checkpoint averaging of (5); (7) continued fine-tuning on Italian NUTSHELL data with checkpoint averaging; (8) (7) with SHAS; (9) chain-of-thought prompting; (10) cascaded system with fine-tuning.

Table A10: MCIF-long mixed-prompt evaluation results across tasks (SQA, SSUM, ASR, ST) and languages.

Method	MC (EN)↑ Acc	EN (ACHAP)		ZH		DE		IT	
		F1	BERT	F1	BERT	F1	BERT	F1	BERT
Baseline	0.00	1.20	86.60	—	—	—	—	—	—
(6) N2+Avg	81.53	35.86	87.84	10.11	62.71	45.46	69.58	40.21	69.37
(7) N2+IT+Avg	81.83	34.39	86.75	10.63	62.15	44.94	69.95	39.11	69.84

Table A11: MC accuracy (EN only) and ACHAP performance across languages. For ACHAP, we report F1 and BERTScore (GC). The baseline is only evaluated on English due to poor performance on other languages. Best values per column are in **bold**.